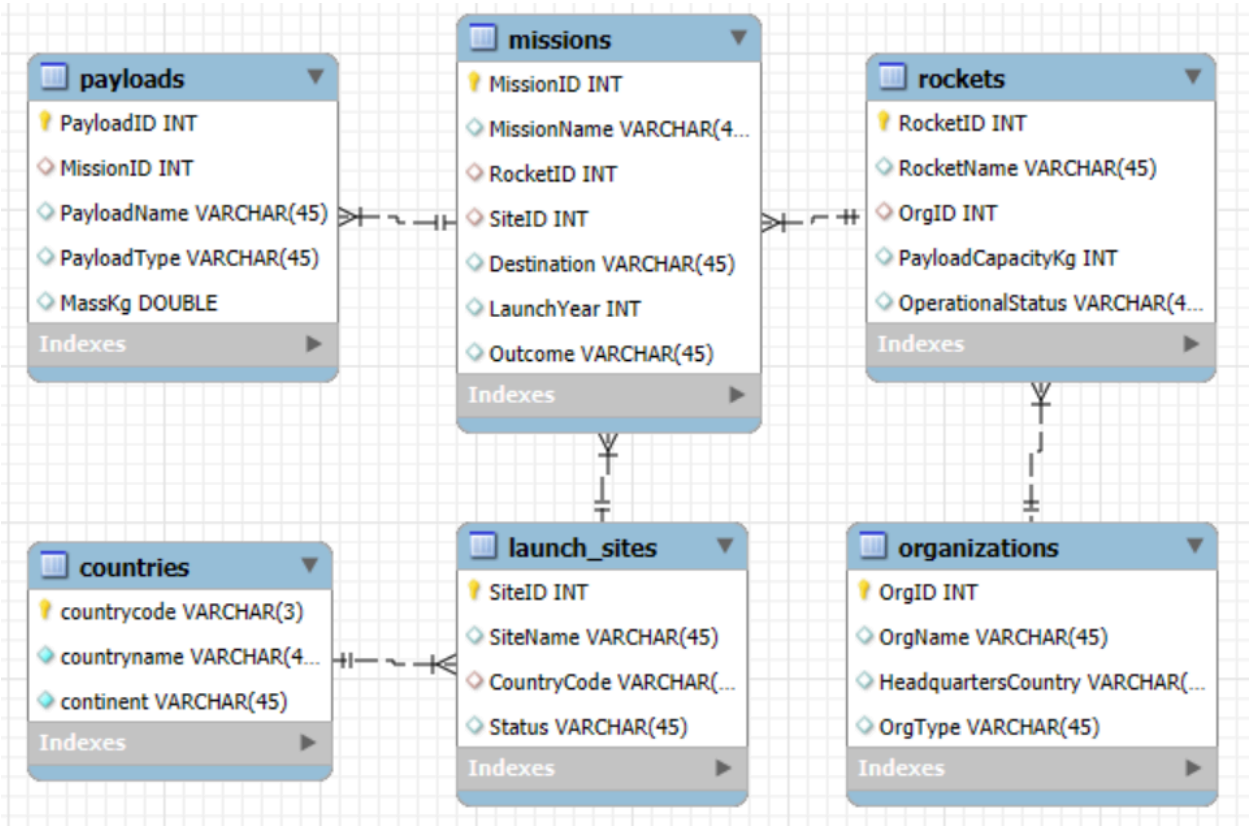
Title: Database Assignment 4
Your Name: Jack Saunders
Date: 10/26/2026



1. Retrieve the mission name, rocket name, organization name, launch site name, destination, launch year, and outcome for all space missions without using table aliases. Sort the results in descending order by launch year and ascending order by mission name.

```
select missions.MissionName,
missions.Destination,
missions.LaunchYear,
missions.outcome,
launch_sites.SiteName,
organizations.Orgname,
rockets.RocketName
```

MissionName	Destination	LaunchYear	outcome	SiteName	Orgname	RocketName
Artemis I	Moon	2024	Success	Kennedy Space Center	National Aeronautics and Space Administration	Saturn V
Chang'e 6	Moon	2024	Success	Jiuquan Satellite Launch Center	China National Space Administration	Long March
Crew-8	ISS	2024	Success	Kennedy Space Center	Space Exploration Technologies Corp	Falcon 9
Gaganyaan Test	LEO	2024	Success	Satish Dhawan Space Centre	Indian Space Research Organisation	PSLV
HTV-X1 Test	ISS	2024	Success	Tanegashima Space Center	Japan Aerospace Exploration Agency	H-IIB
Progress MS-27	ISS	2024	Success	Baikonur Cosmodrome	Roscosmos State Corporation	Proton-M
Sentinel-6B	LEO	2024	Success	Vandenberg Space Force Base	Space Exploration Technologies Corp	Falcon 9
StriX-1	LEO	2024	Success	Mahia Launch Complex 1	Rocket Lab	Electron

```
from missions
join rockets on missions.RocketID =
rockets.RocketID
join organizations on rockets.OrgID =
organizations.OrgID
join launch_sites on
launch_sites.SiteID = missions.SiteID
order by LaunchYear desc,
MissionName asc;
```

2. Retrieve the mission name, rocket name, payload name, payload type, and payload mass (in kg) for all payloads launched in 2024 without using table aliases. Sort the results in descending order by payload mass.

```
select missions.MissionName,
rockets.RocketName,
payloads.PayloadName,
payloads.PayloadType,
payloads.MassKg
from payloads
join missions on
payloads.MissionID =
missions.MissionID
join rockets on missions.RocketID =
rockets.RocketID
where missions.LaunchYear = 2024
order by payloads.MassKg desc;
```

MissionName	RocketName	PayloadName	PayloadType	MassKg
Artemis I	Saturn V	Orion MPCV	Crew Capsule	25800
Crew-8	Falcon 9	Dragon Endeavour	Crew Capsule	12500
HTV-X1 Test	H-IIB	HTV-X Cargo Transporter	Cargo	6000
Progress MS-27	Proton-M	Progress Cargo Supply	Cargo	2450
Chang'e 6	Long March	Queqiao-2 Relay	Sattelite	1200
Sentinel-6B	Falcon 9	Sentinel-6B Altimetry Satellite	Sattelite	1192
Gaganyaan Test	PSLV	Vyommitra Prototype	Robotics	150
StriX-1	Electron	StriX-1 Synthetic Aperture Radar	Sattelite	100

3. Retrieve all launch sites along with their matching mission names, rockets, and launch years for missions that took place in 2024 without using table aliases, ensuring that launch sites with no recorded 2024 missions still appear in the results with null values for the mission fields. Sort the results in alphabetical order by launch site name.

```
select launch_sites.sitename,
missions.missionname,
rockets.rocketname,
missions.launchyear
from launch_sites
left join missions on
launch_sites.siteid =
missions.siteid
    and missions.launchyear = 2024
left join rockets on
missions.rocketid =
rockets.rocketid
order by launch_sites.sitename
asc;
```

sitename	missionname	rocketname	launchyear
Baikonur Cosmodrome	Progress MS-27	Proton-M	2024
Guiana Space Centre	NULL	NULL	NULL
Jiuquan Satellite Launch Center	Chang'e 6	Long March	2024
Kennedy Space Center	Artemis I	Saturn V	2024
Kennedy Space Center	Crew-8	Falcon 9	2024
Mahia Launch Complex 1	StriX-1	Electron	2024
Satish Dhawan Space Centre	Gaganyaan Test	PSLV	2024
Tanegashima Space Center	HTV-X1 Test	H-IIB	2024
Vandenberg Space Force Base	Sentinel-6B	Falcon 9	2024

4. Write a SQL query using a Common Table Expression (CTE) to calculate the total payload mass (in kg) deployed by each organization across all missions without using table aliases. Retrieve the organization name, headquarters country, organization type, and total payload mass, and sort the results in descending order by total payload mass.

```
with Org_Payload_Summary as (
    select rockets.OrgID,
SUM(payloads.MassKg) as
Total_Payload_Mass
    from payloads
    join missions on payloads.MissionID =
missions.MissionID
    join rockets on missions.RocketID =
rockets.RocketID
    group by rockets.OrgID
)
select organizations.OrgName,
organizations.HeadquartersCountry,
```

OrgName	HeadquartersCountry	OrgType	Total_Payload_Mass
National Aeronautics and Space Administration	USA	Government	25800
Space Exploration Technologies Corp	USA	Private	13692
Japan Aerospace Exploration Agency	JPN	Government	6000
Roscosmos State Corporation	RUS	Government	2450
China National Space Administration	CHN	Government	1200
Indian Space Research Organisation	IND	Government	150
Rocket Lab	NZL	Private	100
Arianespace	FRA	Private	NULL

```
    organizations.OrgType,
org_payload_summary.Total_Payload_
Mass
from organizations left join
Org_Payload_Summary
    on organizations.OrgID =
Org_Payload_Summary.OrgID
order by Total_Payload_Mass desc;
```

5. Write a SQL query using two separate Common Table Expressions (CTEs)—one to calculate total missions by country code and another to calculate total payload mass by country code without using table aliases. Then, combine these CTEs with the countries table to display the country name, total number of missions, and total payload mass (in kg). Sort the results in descending order by total payload mass.

```
with mission_counts as (select
launch_sites.countrycode,
        COUNT(missions.missionid) as
total_missions
    from launch_sites
    join missions on launch_sites.siteid =
missions.siteid
    group by launch_sites.countrycode
),
payload_masses as (select
launch_sites.countrycode,
        SUM(payloads.masskg) as
total_payload_mass
    from launch_sites
    join missions on launch_sites.siteid =
missions.siteid
    join payloads on missions.missionid =
payloads.missionid
```

countryname	total_missions	total_payload_mass
United States	3	39492
Japan	1	6000
Russia	1	2450
China	1	1200
India	1	150
New Zealand	1	100
France	0	0
United Kingdom	0	0

```
    group by launch_sites.countrycode
)
select countries.countryname,
coalesce(mission_counts.total_missions
, 0) as total_missions,
coalesce(payload_masses.total_payloa
d_mass, 0) as total_payload_mass
from countries
left join mission_counts on
countries.countrycode =
mission_counts.countrycode
left join payload_masses on
countries.countrycode =
payload_masses.countrycode
order by total_missions desc,
total_payload_mass desc;
```